

Replication of Rubbo (2023) and Oil Cost-Push Markup Shocks

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Abstract

This note documents the current replication and extension of [Rubbo \[2023\]](#). The baseline environment is a multisector New Keynesian model with input-output linkages, sectoral price rigidities, and a divine-coincidence inflation index. We add an oil-sector desired-markup shock, interpreted as a cost-push disturbance, and compare simple monetary-policy responses. The current U.S. calibration shows that stabilizing consumer-price inflation after an oil cost-push shock requires a much larger output-gap contraction than stabilizing the divine-coincidence index. Translating the required output gaps through the IS curve gives a direct measure of the policy-rate movement relative to the natural rate.

Keywords: network Phillips curve; cost-push shock; oil prices; monetary policy; divine coincidence.

JEL Codes: E31; E32; E52; C67.

1 Introduction

[Rubbo \[2023\]](#) revisits the New Keynesian framework in a multisector economy with input-output linkages. The paper shows that sectoral production networks change both the slope of Phillips curves and the inflation index that is most directly connected to welfare. In particular, the divine-coincidence (DC) index summarizes the inflation object that should be stabilized when productivity shocks are the relevant disturbances.

Our extension asks a narrower policy question: how should the central bank react to an oil-price shock when the shock is interpreted as an exogenous increase in desired markups, rather than as a productivity disturbance? This interpretation is useful for oil because the shock acts like a cost-push disturbance that propagates through production chains. It also creates an inflation-output tradeoff: stabilizing CPI inflation is possible, but it may require a large negative output gap and can move DC inflation sharply in the opposite direction.

The note proceeds as follows. Section 2 summarizes the relevant Rubbo objects. Section 3 introduces desired-markup shocks and the static pass-through formulas. Section 4 describes the dynamic policy rules and the interest-rate implementation. Section 5 reports the current U.S. calibration results. Appendix A keeps the replication figures from the original exercise.

2 Model Environment

The economy has N industries indexed by $i, j \in \{1, \dots, N\}$. The calibrated primitives are the labor-share vector α , the final-consumption weight vector β , the input-output matrix Ω , and the diagonal matrix of Calvo price adjustment probabilities $\Delta = \text{diag}(\delta_i)$. As in Rubbo [2023], the Domar exposure vector is

$$\lambda' = \beta'(I - \Omega)^{-1}. \quad (1)$$

Define the rigidity-adjusted Leontief inverse and the common denominator as

$$R_\Delta = (I - \Omega\Delta)^{-1}, \quad d = 1 - \beta'\Delta R_\Delta \alpha. \quad (2)$$

Rubbo's Matlab routines then construct the sectoral Phillips-curve slope vector

$$b = (\gamma + \varphi) \frac{\Delta R_\Delta \alpha}{d}, \quad \kappa^C = \beta' b, \quad (3)$$

where γ is the inverse intertemporal elasticity parameter and φ is the inverse Frisch elasticity parameter. The consumer-price Phillips-curve slope is κ^C .

The sectoral Phillips curves can be written as

$$\pi_t = \rho(I - V)\mathbb{E}_t \pi_{t+1} + b \tilde{y}_t - V \chi_t, \quad (4)$$

where V is the sectoral network matrix, χ_t contains productivity-driven residual terms, and $\tilde{y}_t$ is the output gap. The corresponding household Euler equation is

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma} (i_{t+1} - \mathbb{E}_t \pi_{t+1}^C - r_{t+1}^{nat}). \quad (5)$$

This equation is the bridge from an output-gap path to a nominal policy-rate path.

3 Oil Markup Cost-Push Shocks

Following Rubbo's supplement, we introduce an exogenous desired-markup shock

$$\boldsymbol{\mu}_t^D \in \mathbb{R}^N. \quad (6)$$

A positive entry is a cost-push shock in that sector. A targeted subsidy can be represented as a negative markup shock, so that

$$\boldsymbol{\mu}_t^{D,\text{net}} = \boldsymbol{\mu}_t^{D,\text{shock}} - \mathbf{s}_t. \quad (7)$$

The static CPI markup term computed by the Matlab module is

$$v_t^C = \frac{\beta' \Delta (I - \Omega \Delta)^{-1}}{1 - \beta' \Delta (I - \Omega \Delta)^{-1} \alpha} \boldsymbol{\mu}_t^D. \quad (8)$$

The static DC markup term is

$$\pi_t^{DC} = \rho \mathbb{E}_t \pi_{t+1}^{DC} + (\gamma + \varphi) \tilde{y}_t + \lambda' \boldsymbol{\mu}_t^D. \quad (9)$$

Thus, on impact and abstracting from expectations, the output gap that stabilizes the DC index is

$$\tilde{y}_t^{DC=0} = -\frac{\lambda' \boldsymbol{\mu}_t^D}{\gamma + \varphi}. \quad (10)$$

The dynamic sectoral Phillips curve used in the cost-push module is

$$\boldsymbol{\pi}_t = \rho(I - V) \mathbb{E}_t \boldsymbol{\pi}_{t+1} + b \tilde{y}_t + (I - V) \Delta (I - \Delta)^{-1} \boldsymbol{\mu}_t^D, \quad (11)$$

where the productivity term is set to zero. The oil experiment assumes an AR(1) desired-markup process,

$$\boldsymbol{\mu}_{t+1}^D = \rho_\mu \boldsymbol{\mu}_t^D. \quad (12)$$

The Matlab code solves (11) backward over a finite horizon with terminal condition $\boldsymbol{\pi}_{T+1} = 0$, and reports

$$\pi_t^C = \beta' \boldsymbol{\pi}_t, \quad \pi_t^{DC} = \rho \pi_{t+1}^{DC} + (\gamma + \varphi) \tilde{y}_t + \lambda' \boldsymbol{\mu}_t^D. \quad (13)$$

4 Policy Rules and Interest Rates

The policy exercise expresses monetary policy as a path for the output gap. This is a reduced-form implementation step: first choose the inflation object the central bank wants

to stabilize, then compute the output-gap path required to implement that choice, and finally translate the output-gap path into an interest-rate path through (5).

The rules currently compared are:

$$\tilde{y}_t^0 = 0, \quad (14)$$

$$\tilde{y}_t^{DC} = -\frac{\lambda' \boldsymbol{\mu}_t^D}{\gamma + \varphi}, \quad (15)$$

$$\tilde{y}_t^{lean,CPI} = \theta_C \tilde{y}_t^{CPI}, \quad (16)$$

$$\tilde{y}_t^{lean,DC} = \theta_{DC} \tilde{y}_t^{DC}. \quad (17)$$

The CPI-stabilizing rule chooses $\tilde{y}_t$ so that $\beta' \boldsymbol{\pi}_t = 0$ in the backward recursion:

$$\tilde{y}_t^{CPI} = -\frac{\rho \beta' (I - V) \boldsymbol{\pi}_{t+1} + \beta' (I - V) \Delta (I - \Delta)^{-1} \boldsymbol{\mu}_t^D}{\beta' b}. \quad (18)$$

The transparent comparison loss is quadratic:

$$\mathcal{L}^{CPI} = \sum_{t=0}^{T-1} \rho_L^t [(\pi_t^C)^2 + \omega_y \tilde{y}_t^2], \quad (19)$$

$$\mathcal{L}^{DC} = \sum_{t=0}^{T-1} \rho_L^t [(\pi_t^{DC})^2 + \omega_y \tilde{y}_t^2], \quad (20)$$

$$\mathcal{L}^{dual} = \sum_{t=0}^{T-1} \rho_L^t [\omega_C (\pi_t^C)^2 + \omega_{DC} (\pi_t^{DC})^2 + \omega_y \tilde{y}_t^2]. \quad (21)$$

These losses are diagnostic statistics, not the full second-order welfare matrix in [Rubbo \[2023\]](#).

Rearranging the IS curve gives the policy-rate movement relative to the natural rate:

$$i_{t+1} - r_{t+1}^{nat} = \mathbb{E}_t \pi_{t+1}^C + \gamma (\mathbb{E}_t \tilde{y}_{t+1} - \tilde{y}_t). \quad (22)$$

The code uses the simulated future paths as model-consistent expectations, with terminal values set to zero:

$$\hat{i}_{t+1} \equiv i_{t+1} - r_{t+1}^{nat} = \pi_{t+1}^C + \gamma (\tilde{y}_{t+1} - \tilde{y}_t). \quad (23)$$

Because the exercise is a desired-markup shock rather than a productivity shock, the reported $\hat{i}_{t+1}$ is the policy-rate response above the natural-rate path.

5 Results

The current baseline is a 10 percent desired-markup shock in the oil sector with $\rho_\mu = 0.80$, using the U.S. 2012 calibration from the replication files. Table 1 reproduces the oil productivity-shock benchmark, while Tables 2 and 3 report the new cost-push markup calculations.

The distinction between the productivity-shock benchmark and the markup-shock exercise is important. In the original oil experiment, the disturbance changes the productive capacity of the oil sector and therefore enters the Phillips curves through the productivity residuals. In the extension, the disturbance raises desired markups and is therefore closer to a cost-push shock: firms would like to charge a higher relative price even without a change in physical productivity. This is the relevant experiment for asking how monetary policy should react to an adverse oil-price movement that looks like a supply-driven increase in marginal costs.

Table 1: Consumer inflation after a 10% negative oil-sector productivity shock

	Full model		
	$\delta = \text{actual}$	$\delta \equiv \delta_{\text{mean}}, \delta_{\text{oil}} = 1$	$\delta \equiv \delta_{\text{mean}}$
Sticky wages	0.22	0.07	-0.00
Flexible wages	0.18	0.01	-0.06

Notes: Values are percentage points. This table is the original oil-shock benchmark from the Rubbo replication code.

Table 1 is a useful benchmark because it shows how much of the original oil result comes from the joint role of price rigidity, wage rigidity, and input-output propagation. With actual sectoral price-adjustment probabilities, a 10 percent negative oil-sector productivity shock raises consumer inflation by 0.22 percentage points under sticky wages and by 0.18 percentage points under flexible wages. The sticky-wage case is larger because wage rigidity prevents part of the adjustment from occurring through wages, so more of the disturbance appears in goods prices. When all sectors are assigned the mean price flexibility but the oil sector is kept fully flexible, the response falls to 0.07 and 0.01 percentage points. When all sectors receive the mean value of δ , the CPI effect is essentially zero under sticky wages and slightly negative under flexible wages. The table therefore says that oil is not important only because it is an expenditure item; it matters because oil is located in the production network and because its price rigidity differs from the average sector.

Table 2 repeats the oil experiment but changes the nature of the shock. The markup

Table 2: CPI response to a 10% oil-sector markup shock

	$\delta = \text{actual}$	$\delta \equiv \delta_{\text{mean}}, \delta_{\text{oil}} = 1$	$\delta \equiv \delta_{\text{mean}}$
CPI response	0.242	0.084	0.016

Notes: Values are percentage points. The underlying Matlab outputs are 0.0024183, 0.00083976, and 0.00015828, respectively.

shock produces a 0.242 percentage-point CPI response under the actual calibration, slightly larger than the sticky-wage productivity benchmark in Table 1. This happens because the desired-markup shock enters directly as a cost-push term: it is not mediated by productivity and therefore creates inflationary pressure even when output is held fixed. The alternative δ experiments again show that sectoral price rigidity is central. Setting all sectors to the mean δ while making oil fully flexible lowers the CPI response to 0.084 percentage points, and setting all sectors to the mean δ lowers it further to 0.016 percentage points. The sharp decline indicates that the calibrated heterogeneity in adjustment probabilities is doing substantial work in the static pass-through calculation.

Table 3: CPI, divine-coincidence inflation, and output-gap response

	CPI markup	DC markup	Output gap stabilizing DC
Response	0.002418	0.006851	−0.002284

Notes: Values are decimal units. In percentage points, the responses are 0.242, 0.685, and −0.228, respectively.

Table 3 explains why the monetary-policy problem is not well summarized by CPI alone. The CPI markup term is 0.002418 in decimals, or 0.242 percentage points. The DC markup term is 0.006851, or 0.685 percentage points, almost three times as large. This gap arises because the DC index weights sectors by their network relevance for the divine-coincidence condition, while CPI weights sectors by final-consumption exposure. Oil has a larger role as an upstream input than as a direct final-consumption item, so the network-based inflation pressure is larger than the consumer-price pressure. The output gap that stabilizes DC inflation on impact is −0.002284, or −0.228 percentage points. This number is the first measure of the real activity cost of stabilization: keeping the network-relevant inflation object stable requires a contraction, but a relatively modest one compared with the CPI-stabilizing rule shown below.

Figure 1 shows the dynamic inflation responses to the oil markup shock. CPI inflation rises on impact and gradually decays as the AR(1) markup shock dies out. The DC response

is much larger on impact and also more persistent in levels, reflecting the fact that the shock travels through the input-output network before reaching final prices. This is the dynamic counterpart of Table 3: CPI is the object consumers see directly, but DC is the object that internalizes the upstream pressure created by oil as an intermediate input. The figure also shows the illustrative subsidy experiment. Offsetting part of the oil-sector markup reduces the CPI response at every horizon, but it does not change the qualitative propagation mechanism: the shock still fades gradually because the underlying markup process is persistent.

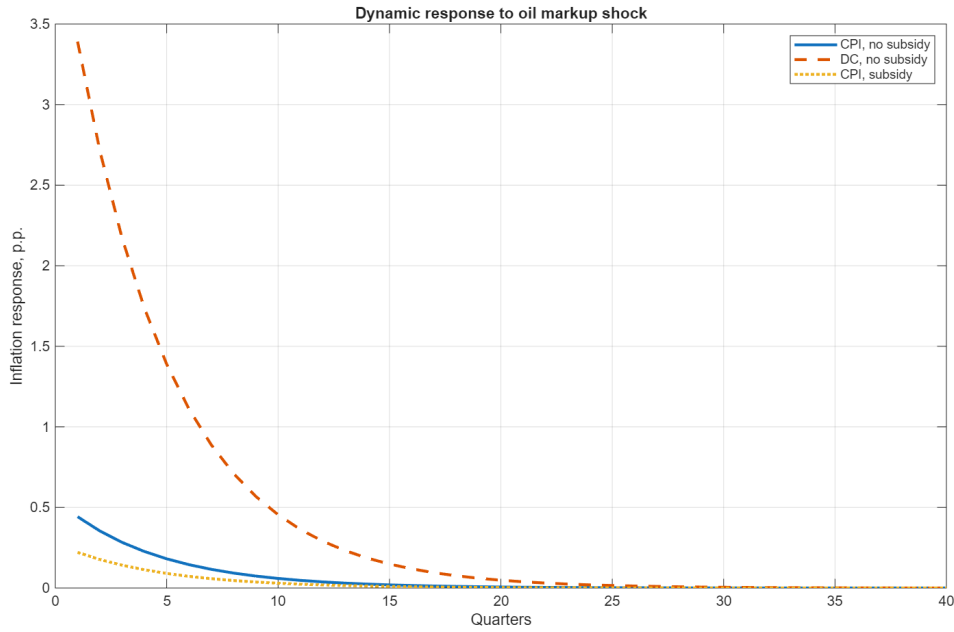


Figure 1: Dynamic inflation response to an oil-sector markup shock

Figure 2 summarizes the network-pressure object generated by the sectoral Phillips-curve markup term. The no-subsidy line is the chain-wide pressure created by the oil cost-push shock. Its impact value is much larger than the CPI response because this object aggregates sectoral pressure before final-consumption weights compress it. The subsidy line shows how a targeted offset reduces propagated pressure along the production chain, not merely the direct oil-sector price. This is important for policy design: a sectoral fiscal instrument can dampen the source of the network shock, while monetary policy can only trade off inflation stabilization against the output gap.

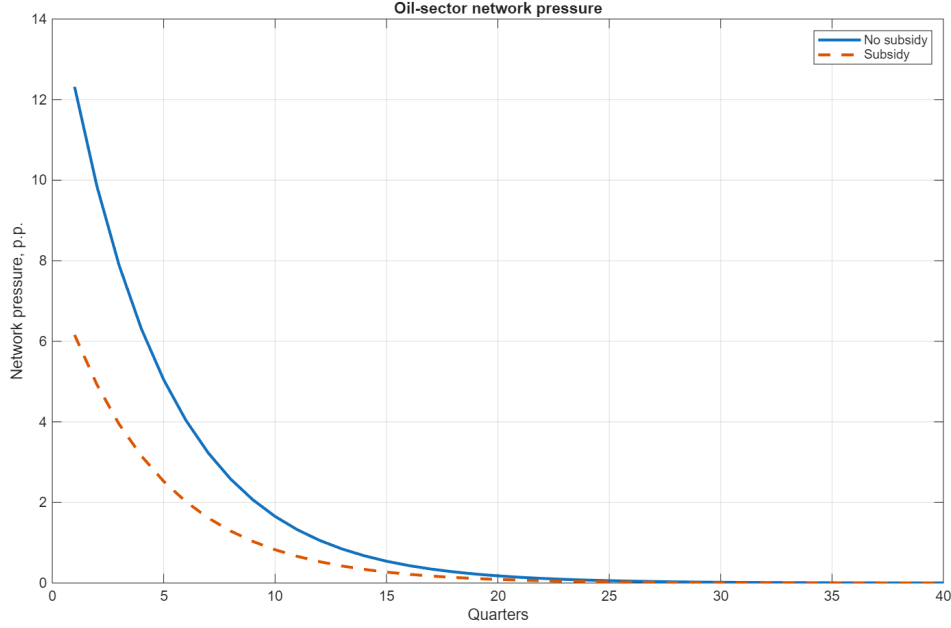


Figure 2: Network pressure from the oil-sector markup shock

5.1 Monetary-Policy Comparison

Figures 3–6 compare the five policy rules. The rules should be read as different answers to the same central-bank question. The zero-gap rule asks what inflation does if the central bank does not move real activity in response to the oil markup shock. The CPI-stabilizing rule asks how contractionary policy must be to keep consumer-price inflation at zero. The DC-stabilizing rule asks how much policy must move output to stabilize the network-relevant inflation index. The two lean-against rules are partial responses, halfway between accommodation and exact stabilization.

Figure 3 focuses on consumer-price inflation. By construction, the CPI-stabilizing rule keeps the CPI response at zero. The zero-gap rule produces the largest CPI response, because monetary policy does not create any offsetting demand contraction. The DC-stabilizing rule still lowers CPI inflation relative to the zero-gap benchmark, but it does not eliminate it. This is the main policy tradeoff: stabilizing the inflation index that is most closely connected to the production network allows some temporary movement in the consumer price index.

Figure 4 reverses the ranking. The DC-stabilizing rule sets the divine-coincidence response to zero, while the CPI-stabilizing rule generates a large negative DC response. The reason is mechanical but economically revealing: to eliminate CPI inflation after an oil cost-push shock, policy must engineer a sizeable negative output gap. That contraction more than offsets the DC markup term, so the DC index undershoots. In other words, strict CPI

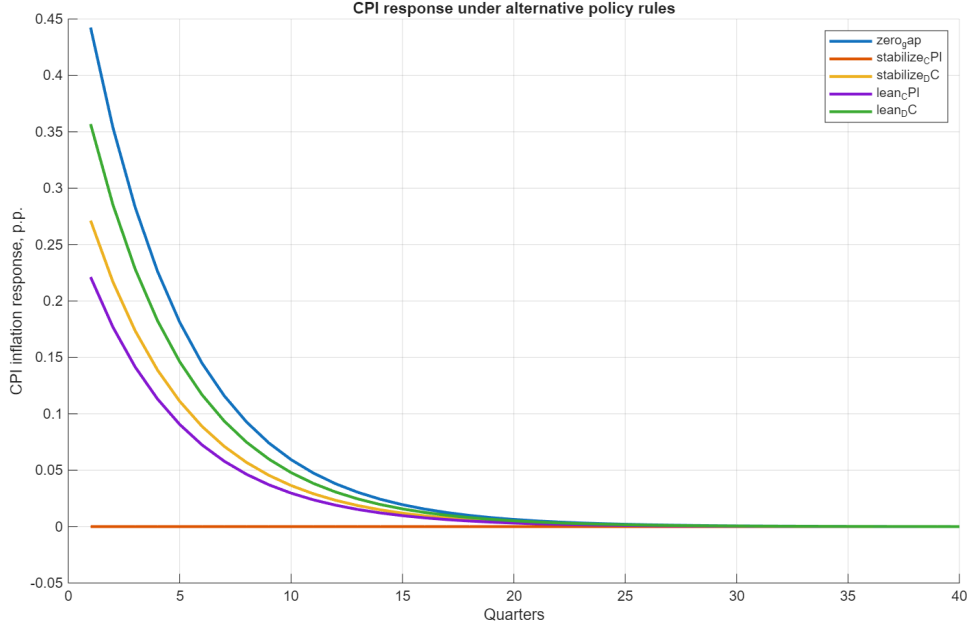


Figure 3: CPI response under alternative monetary-policy rules

stabilization may look successful from the perspective of headline inflation while creating an excessive contraction relative to the network-based stabilization criterion.

Figure 5 makes the real-side cost explicit. The zero-gap rule has no output response by construction. The CPI-stabilizing rule requires the deepest contraction, with an impact output gap close to -0.59 percentage points. The DC-stabilizing rule requires a smaller contraction, around -0.228 percentage points on impact. The partial rules sit between these extremes. This figure is especially useful for interpreting the policy problem: the central bank can always reduce inflation by contracting demand, but the relevant question is whether the additional inflation stabilization is worth the additional output loss.

Figure 6 summarizes these tradeoffs in the diagnostic quadratic losses. A CPI-only loss rewards the CPI-stabilizing rule for eliminating headline inflation, but still penalizes the output contraction required to do so. A DC loss penalizes the CPI-stabilizing rule heavily because that rule creates a large DC undershooting. Under the dual criterion, which cares about both CPI and DC inflation as well as the output gap, the DC-stabilizing rule performs best in this calibration. The result should not be interpreted as a full welfare ranking yet, since the loss here is intentionally transparent and not Rubbo's full welfare matrix. It is nevertheless a clear diagnostic: a central bank that reacts only to CPI may over-tighten after an oil cost-push shock.

Table 4 reports the same comparison numerically. The zero-gap rule has no output cost,

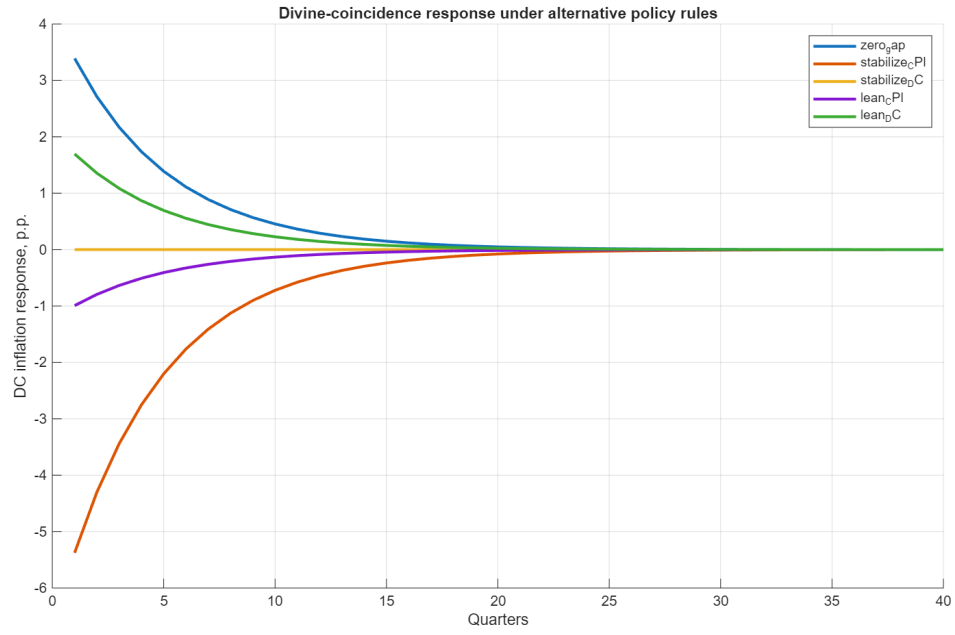


Figure 4: Divine-coincidence inflation response under alternative monetary-policy rules

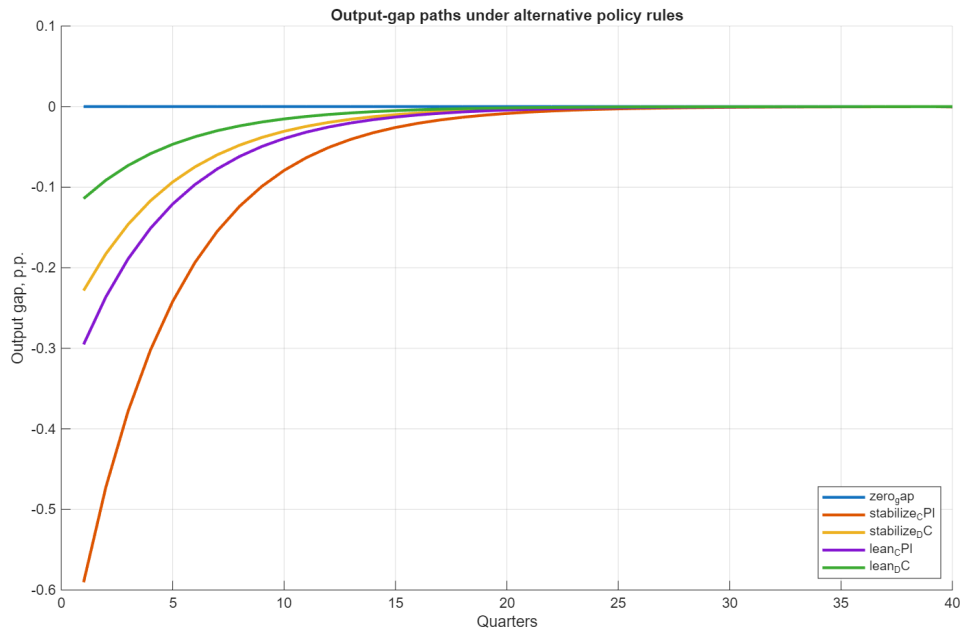


Figure 5: Output-gap paths under alternative monetary-policy rules

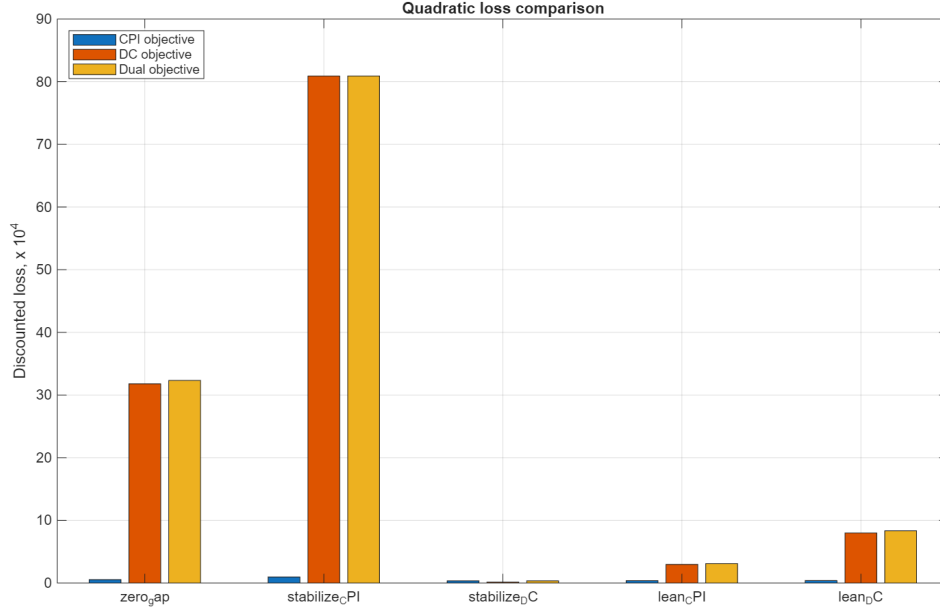


Figure 6: Quadratic loss comparison under alternative monetary-policy rules

but it allows a peak CPI response of 0.442 percentage points and a peak DC response of 3.391 percentage points. Its total loss is therefore dominated by DC inflation. The strict CPI rule eliminates CPI inflation, but it requires the largest output movement, 0.590 percentage points in absolute value, and generates the largest DC deviation, 5.377 percentage points. This is why its DC and total losses are the highest in the table. The strict DC rule sets the peak DC response to zero, leaves a peak CPI response of 0.271 percentage points, and requires a smaller output gap of 0.228 percentage points. It therefore minimizes the DC loss and the total loss in the current diagnostic criterion. The two lean-against rules show the value of partial accommodation: leaning against CPI lowers CPI inflation relative to zero-gap policy but still keeps DC deviations limited, while leaning against DC uses the smallest nonzero output response but leaves more inflation volatility.

Table 5 converts the output-gap rules into policy-rate movements relative to the natural rate using (23). These responses should be interpreted carefully. They are not policy-rate levels and they are not deviations from the pre-shock nominal rate. They are the nominal rate movements above the natural-rate path needed to implement each output-gap rule through the IS curve. Because the experiment is a pure markup shock, the productivity component of the natural rate is unchanged in the current implementation.

The zero-gap rule requires an impact rate increase of 0.354 quarterly percentage points, or 1.416 annualized percentage points. This is the rate movement that prevents the output gap

Table 4: Summary statistics under alternative monetary-policy rules

Policy rule	Peak $ \pi^{CPI} $	Peak $ \pi^{DC} $	Max $ \tilde{y} $	Loss CPI $\times 10^4$	Loss DC $\times 10^4$	Total loss $\times 10^4$
Zero output gap	0.442	3.391	0.000	0.541	31.792	32.334
Stabilize CPI	0.000	5.377	0.590	0.964	80.907	80.907
Stabilize DC	0.271	0.000	0.228	0.348	0.144	0.348
Lean against CPI	0.221	0.993	0.295	0.376	2.968	3.103
Lean against DC	0.357	1.696	0.114	0.388	7.984	8.336

Notes: Inflation and output-gap responses are percentage points. Losses are reported in units of 10^{-4} . Numerical zeros correspond to values close to machine precision.

from moving despite higher expected CPI inflation. The strict CPI-stabilizing rule requires a smaller rate-gap movement, 0.118 quarterly percentage points or 0.472 annualized percentage points, because the rule deliberately allows a large contemporaneous negative output gap. The interest-rate formula contains $\gamma(\tilde{y}_{t+1} - \tilde{y}_t)$; when policy creates a sharp negative output gap on impact that subsequently closes, this expected recovery partly substitutes for a larger rate increase. The DC-stabilizing rule requires 0.263 quarterly percentage points, or 1.051 annualized percentage points. It is therefore more contractionary in interest-rate space than strict CPI stabilization, but much less contractionary in output-gap space. This distinction is central: the severity of a policy rule cannot be read from the interest-rate response alone; it must be read jointly with the output-gap path.

The two partial rules make the same point. Leaning against CPI requires an annualized rate response of 0.944 percentage points and a maximum output gap of 0.295 percentage points. Leaning against DC requires a larger annualized rate response, 1.233 percentage points, but a smaller output-gap movement of 0.114 percentage points. The ranking differs because the IS curve maps rates into the change in the output gap, not only into the level of the output gap. For the current calibration, the preferred rule under the diagnostic loss is not the rule with the smallest rate response; it is the rule that best balances CPI inflation, DC inflation, and output stabilization.

6 Conclusion

The current exercise suggests that the answer to the policy question depends on which inflation index the central bank treats as the relevant target after an oil cost-push shock. If it insists on stabilizing CPI exactly, the model requires a large output-gap contraction and generates a severe movement in DC inflation. If it stabilizes the DC index, the output contraction is smaller and the diagnostic loss is lower, while CPI inflation is allowed to move

Table 5: Interest-rate responses under alternative monetary-policy rules

Policy rule	Impact, q. p.p.	Peak $ q $, p.p.	Impact, ann. p.p.	Peak $ ann. $, p.p.
Zero output gap	0.354	0.354	1.416	1.416
Stabilize CPI	0.118	0.118	0.472	0.472
Stabilize DC	0.263	0.263	1.051	1.051
Lean against CPI	0.236	0.236	0.944	0.944
Lean against DC	0.308	0.308	1.233	1.233

Notes: Quarterly responses are percentage points. Annualized responses multiply the quarterly rate by four.

temporarily. The next step is to connect this reduced-form policy comparison to Rubbo's full welfare matrix and then rebuild the primitives α , β , and Ω with Brazilian input-output data.

References

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A Replication Figures from Rubbo's Baseline

This appendix keeps the figures produced by the baseline replication routines. They are useful checks on the calibrated objects before adding the oil markup-shock layer.

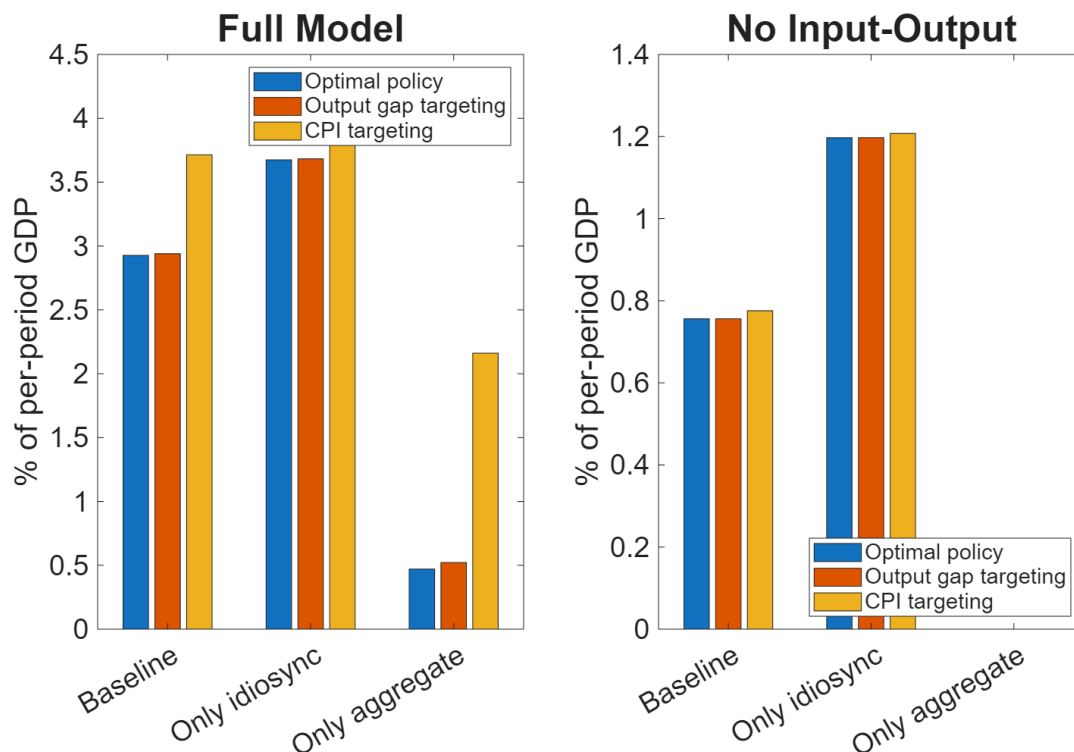


Figure 7: Welfare losses under alternative policy rules in the baseline replication

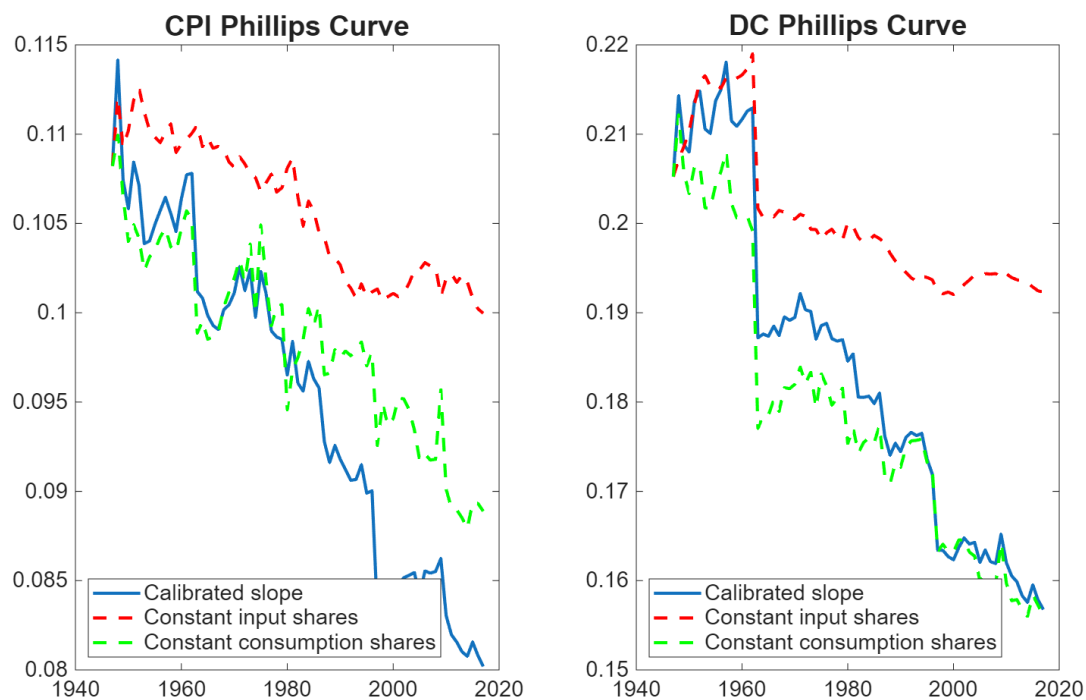


Figure 8: CPI and divine-coincidence Phillips-curve slopes over time

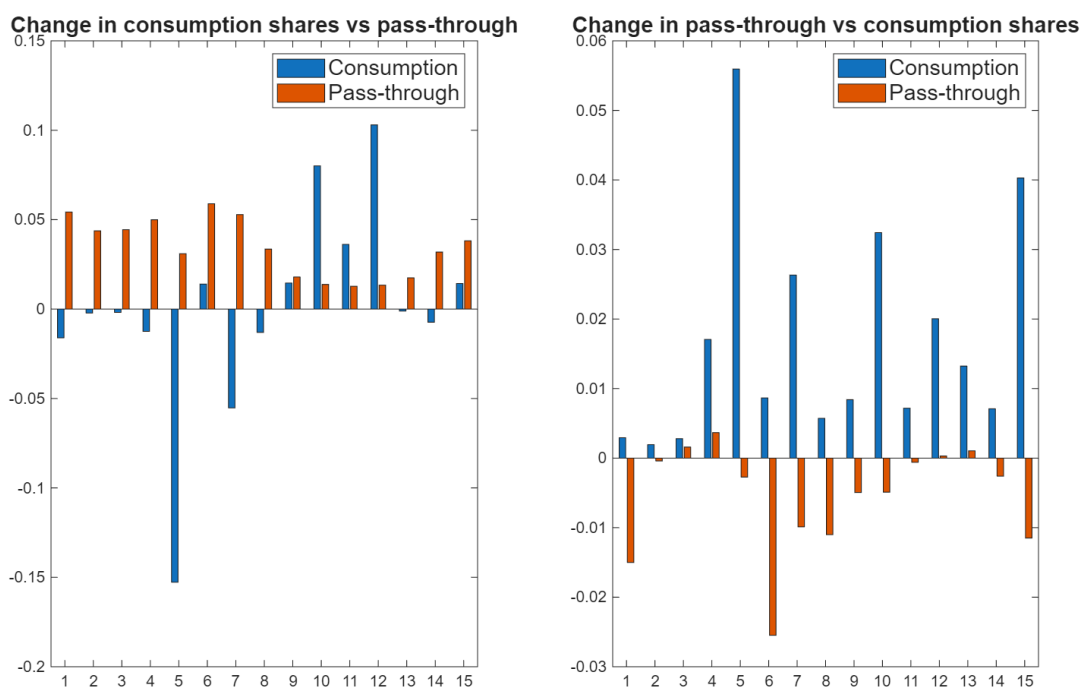


Figure 9: Consumption shares and pass-through changes in the baseline replication

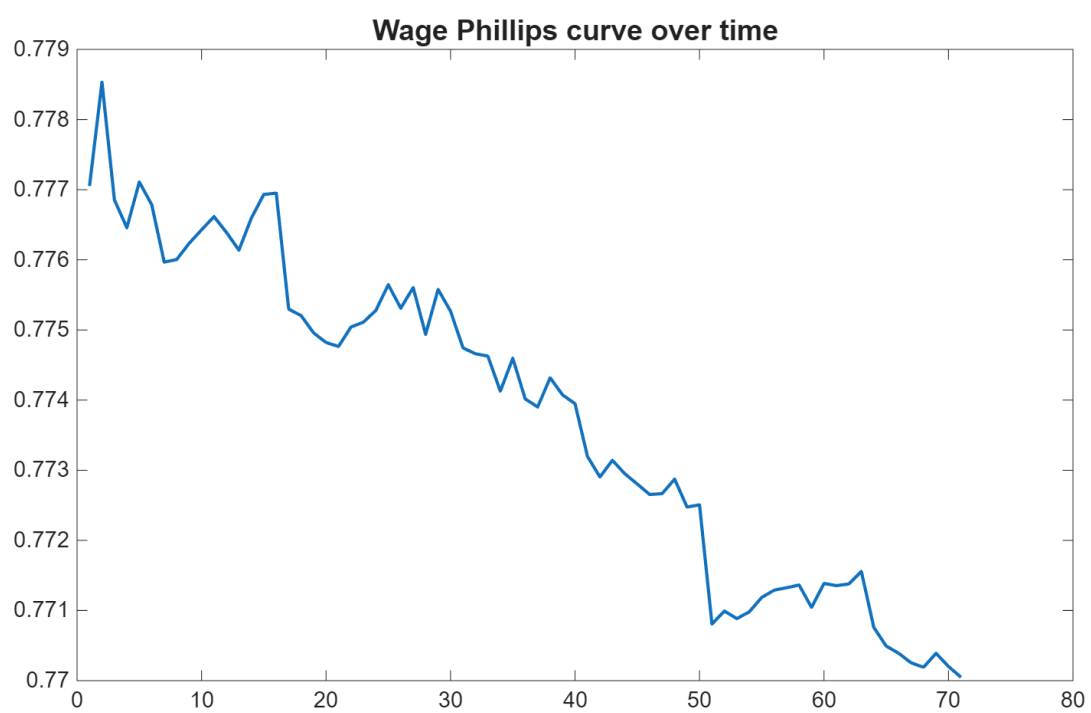


Figure 10: Wage Phillips-curve slope over time in the baseline replication